

Alexander J. Elliott

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Video Editing • Robotics • Guitar • Photography • Physical AI • Climbing • Geography

EXPERIENCE

Magna International • Mechatronics CO-OP (CAE Team)

Jan - Aug 2025 • Newmarket, ON

- Developed internal Python and TypeScript applications to automate pre- and post-processing of FEA, multibody dynamics, and inertial analysis, reducing repetitive engineering effort across global CAE teams
- Architected and deployed Inertia+, a software platform that automated component inertia calculations and engineering report generation, reducing engineering effort by up to 26% (270+ hours saved per quarter)
- Led development of Collab Space, a web application built with JavaScript and Python for interactive FEA visualization, enabling real-time collaboration and faster engineering design reviews

Key Skills: Engineering Software • Python • TypeScript • Simulation Automation • Data Visualization

AME Group • Mechanical CO-OP & QAQC CO-OP

Sept - Dec 2023, May - Aug 2024 • Vancouver, BC

- Automated drafting and engineering workflows using Python, PyAutoCAD, and Revit plugins
- Developed C# automation tools to generate schematics, convert AutoCAD details, and improve productivity
- Led BIM improvement initiatives that standardized engineering workflows and enabled teams to work more efficiently within Revit

Key Skills: Engineering Automation • Python • C# • Revit API • Process Improvement

UBC Uncrewed Aircraft Systems • Mechanical Payload Team

Sep 2022 – Sep 2024 • Vancouver, BC

- Designed and prototyped payload mechanisms using electromagnetic clutches, viscous damping, and guided parachutes
- Designed and manufactured carbon-fibre aircraft structures and integrated payload systems
- Helped achieve 2nd place (Phase 1) and 3rd place (Phase 2) at the 2024 AEAC competition

Key Skills: UAV Systems • Mechanical Design • Payload Integration • Composite Manufacturing • Rapid Prototyping

PROJECTS

Automated Spice Dispensing Robot • UBC Mech 423 Final Project

Jan – Apr 2026

- Designed and developed a robotic dispensing system integrating embedded electronics and desktop software
- Designed embedded motor-control firmware for an MSP430-based control system using IR sensors and limit switches
- Developed a C# interface supporting voice control, automated dispensing, and inventory management

Key Skills: Embedded Systems • Robotics • Motor Control • PCB Design • Sensors • C • C#

Litter Collecting Drone • UBC Mech 458 Capstone

Sep 2025 – Apr 2026

- Designed and manufactured a quadcopter to autonomously detect and retrieve litter in difficult terrain
- Integrated YOLOv8 on a Raspberry Pi Zero 2 W for onboard litter detection
- Programmed a Pixhawk flight controller and integrated perception, flight control, and mechanical systems

Key Skills: Robotics • Computer Vision • YOLOv8 • UAV • Embedded AI • Systems Integration

EDUCATION

University of British Columbia

2021 - 2026 • Vancouver, BC

BASc Mechanical Engineering • Mechatronics Option • w/ Co-Op • w/ Distinction • 5x Dean's List

SKILLS

Robotics: Embedded Systems • Computer Vision • Motor Control • Sensors • UAV Systems

Programming: Python • C • C# • Java • MATLAB

AI & Software: YOLOv8 • OpenCV • TensorFlow • React

Hardware: MSP430 • Raspberry Pi • Arduino • PCB Design • SolidWorks • Onshape